Change Report & Implementation Summary

6-DOF Robotic Arm Pick-and-Place and Color Sorting

Repository: https://github.com/saumyadebug/robotic-arm-pick-and-place

Report source: 24BRS1075_24BRS1065_24BRS1126_Poster_A3 (7).pdf

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1. Purpose of This Work

The project report (poster) defines the target system: a 6-DOF robotic arm actuated by six MG995 servo motors, controlled by an Arduino Uno via a PCA9685 PWM driver, powered by a 7.4 V battery through an XL4016 buck converter, with a gripper-mounted TCS3200 color sensor sorting objects into four bins (Red, Green, Blue, Yellow). The repository did not fully match these requirements, so this work updates the firmware, hardware documentation, and test records to align with the report, and delivers an implementation plan plus this change report.

2. What Was Changed (and Why)

Firmware: integrated_sorting.ino was updated to actuate a 6th joint (wrist roll, channel 5), classify four colors including Yellow, debounce ultrasonic detection, and standardize PCA9685 initialization. pick_and_place.ino, color_sensor_test.ino, and end_effector_test.ino received consistent channel definitions and oscillator setup. Hardware docs were aligned to XL4016, MG995 servos, 6 joints, and a gripper-mounted sensor. Test records now include channel 5 and Yellow cycles. README status updated to Completed (v1.1).

3. What Was Added

- CH_WRIST_ROLL (channel 5) joint support across sketches (6-DOF).
- COLOR_YELLOW enum value and 4th bin target (base tick 590).
- Yellow classification logic in scanColor() and color_sensor_test.
- Ultrasonic debounce (3 consecutive readings) in STATE_IDLE.
- Filter-settling delay (10 ms) between color filter switches.
- Gripper-mounted sensor note in sensor_integration.md.
- XL4016 buck converter specification in components.md / power_supply.md.
- Wrist roll row in servo_test_results.md.
- Yellow cycle row in pick_and_place_test_results.md.
- IMPLEMENTATION_PLAN.md (root and docs/).
- This change-report PDF.

4. What Was Removed

No files were deleted. Unused USMIN/USMAX constants in servo_testing.ino are retained for reference. No functional behavior was removed; only refinements and additions.

5. Per-File Change Details

code/integrated_sorting/integrated_sorting.ino
- Added 6th joint; 4-color FSM; debounce; standardized init; Yellow bin.

code/pick_and_place/pick_and_place.ino
- Added wrist roll channel; standardized init; 6-joint moves.

code/color_sensor_test/color_sensor_test.ino
- Added Yellow classification (R&G strong, B weak).

code/end_effector_test/end_effector_test.ino
- Standardized PCA9685 oscillator frequency.

hardware/components.md
- Voltage regulator changed to XL4016.

hardware/power_supply.md
- Buck converter specified as XL4016.

hardware/pin_connections.md
- Channel 5 defined as Wrist Roll (MG995).

docs/system_architecture.md
- Diagram updated to 6 MG995 joints.

docs/development_log.md
- Updated to 6-DOF assembly.

docs/sensor_integration.md
- Added gripper-mounted sensor note.

tests/servo_test_results.md
- Added channel 5 row.

tests/pick_and_place_test_results.md
- Added Yellow cycle; updated summary.

README.md
- Status -> Completed (v1.1); progress checkboxes updated.

IMPLEMENTATION_PLAN.md
- New implementation plan.

docs/implementation_plan.md
- Copy of implementation plan.

6. Implementation Plan Summary

Phase 1 · Firmware Core: 6-DOF + 4-color FSM (completed).

Phase 2 · Hardware Documentation Alignment (completed).

Phase 3 · Test Records & Status (completed).

Phase 4 · Planning & Reporting Artifacts (completed; optional CI/tests remain).

Phase 5 · Future Improvements (backlog): interpolation, IK, conveyor, benchmarks.

7. Risks & Mitigations

- No Arduino CLI available: changes kept minimal and consistent; manual review performed.
- Yellow misclassification under ambient light: shielding & white-balance planned (Phase 5).
- 6th servo mechanical interference: calibrate via joint_calibration before full runs.
- Base servo range (150-600) for 4 bins: targets spaced 200/375/550/590.

8. Acceptance Criteria

- All sketches consistent (standardized init, channel defs).
- FSM supports 4 colors and 6 joints.
- Documentation matches report (6 MG995, XL4016, gripper-mounted sensor, 4 bins).
- Test records include 6 channels and Yellow.
- Implementation plan and change-report PDF delivered.

9. Deliverables

- Updated repository (firmware + docs + tests).
- IMPLEMENTATION_PLAN.md.
- docs/change_report.pdf (this document).
- Git commit and push to origin/main.